

Passable.

Measuring whether a city is actually accessible — from the desk, at city scale.

A ramp on a map is not a usable ramp.

ADA compliance is a measurement problem — slope, width, clearance. We measure all of it automatically from Cyvl's calibrated imagery and LiDAR, and turn "we have 748 ramps" into "412 of them fail, here's the slope on each."

8.33%

max legal ramp slope — the line between accessible and not, and you can't see it by eye

\$1.4B

L.A.'s curb-ramp settlement (Willits) — the cost of getting accessibility wrong

desk

where the audit happens — no field crew, no measuring wheel, no volunteers

What it is: a gov-tech accessibility-audit tool built on the Cyvl Spatial SDK · measurement, not guesswork · adoptable by a city tomorrow

1 · The problem

Every US city with 50+ employees is legally required (ADA Title II) to keep an **ADA Transition Plan** — a self-evaluation of whether its sidewalks, curb ramps, and crossings are actually usable by people with disabilities. Most cities' plans are decades out of date, because the audit is brutal: a field crew walks every ramp with a digital level and a tape measure, recording slope and width one corner at a time. Somerville's backlog is an estimated **\$75M**, funded at ~\$1M/year — and the city resorted to **volunteers** for its accessibility survey.

The stakes aren't paperwork. A curb ramp at 9% instead of 8.33% is the difference between a wheelchair user crossing independently or being stuck. And the legal exposure is enormous — L.A. settled **Willits v. City of Los Angeles for \$1.4 billion** over curb-ramp accessibility; Boston has its own curb-ramp obligations. Cities are under court-ordered deadlines with no scalable way to measure compliance.

The core insight: ADA compliance is fundamentally a *measurement* regime — running slope $\leq 8.33\%$, cross-slope $\leq 2\%$, clear width $\geq 36\text{--}48"$. You cannot judge it from a photo by eye, and you cannot field-measure a whole city by hand. That gap is the product.

2 · What Passable does

Passable audits the accessibility of the built environment automatically, from data Cyvl already captured. For every curb ramp and sidewalk segment:

1. **Locate** — find the ramp in the street imagery (the SDK's text-prompt segmentation, `locate(frame, "curb ramp")`, lifted to 3D).
2. **Measure** — compute its real-world running slope, cross-slope, and width from the calibrated LiDAR + camera geometry (`measure()` / `pixel→3D unproject()`). This is survey-grade geometry, not a guess.
3. **Verdict** — score each against the ADA/PROWAG thresholds: pass / fail / borderline, with the exact number and the photo as evidence.
4. **Map + plan** — a city-wide accessibility map (green/amber/red), ranked worst-first, exportable as the Transition-Plan work-list.

The output a city has never had: not "we own 748 ramps," but **"412 fail, here is the slope on each, prioritized, with photographic proof."**

Why this is possible now and wasn't last week: the Cyvl platform gives asset *locations* with no condition or geometry. The new **Spatial SDK** exposes calibrated camera poses, LiDAR depth, and pixel↔3D projection — so you can extract real measurements from a single photo. Passable is impossible on the platform alone; it's native to the SDK.

3 · The business

This is a gov-tech (B2G) startup with a parallel private-sector market — both driven by legal mandate, which means non-discretionary budgets and recurring demand.

Buyer 1 — Cities (B2G)

Every Title II entity (50+ employees) must maintain a Transition Plan. They currently pay engineering consultants for manual field surveys or scrape by with volunteers. Passable replaces the survey with software over an existing scan. **Sold as:** per-mile or per-city annual subscription. The buyer is the DPW / accessibility office — or the CIO's innovation team.

Buyer 2 — Private property (B2B)

ADA Title III drives a large drive-by-litigation industry against retail chains, REITs, and property managers. They pay for compliance audits and, critically, **defensible documentation**. **Sold as:** audit-as-a-service per portfolio, plus monitoring.

Why the unit economics work

Path	Cost to audit one ramp	A 5,000-ramp city
Manual field survey (crew + digital level + documentation)	~\$50–150	\$250k–750k, and stale in a year
Passable (software over an existing Cyvl scan)	~\$0 marginal	a software subscription

The margin between those two columns is the business. And it's **recurring**: ramps settle, crack, and get damaged; Transition Plans need periodic re-survey; every time Cyvl rescans a city, Passable re-audits it automatically. Survey work that was a one-off consulting invoice becomes a subscription.

Where it sits in the market

- **Today's solution:** ADA engineering consultants doing manual surveys (firms like those selling Rieker digital-level hardware to field crews). Slow, expensive, point-in-time.
- **The wedge:** nobody audits accessibility *automatically, at city scale, from a street-level 3D scan*. That requires exactly the calibrated-imagery-plus-LiDAR layer Cyvl/the SDK provides.
- **Relationship to Cyvl:** Cyvl scans the streets; Passable is the accessibility-analytics vertical on top — a natural data partnership or platform extension, which is also why it's a strong thing to show *at Cyvl's own event*.

The market in one line: a legally mandated, perpetually-stale, manually-surveyed compliance problem in every city and every commercial property in the country — converted from field labor into software over data that already exists.

4 · The data & the SDK primitives it rides

Capability used	SDK primitive	Role in Passable
Find the ramp in a photo	<code>locate(frame, "curb ramp")</code> (SAM3 → 3D)	Detect & isolate each ramp / sidewalk run
Measure slope & width	<code>unproject()</code> / <code>measure()</code> via LiDAR depth	Real-world running slope, cross-slope, clear width
Evidence photo per asset	<code>nearest_frame()</code> , <code>frame.show()</code>	Proof image attached to each verdict
Known asset starting points	748 ramps + 227k ft sidewalk (feature layer)	Seed locations to measure — no blind search
CAD / 3D export	PLY / point-cloud export	Hand the failing-ramp work-list to the design team (Autodesk)

Thresholds applied: curb-ramp running slope $\leq 8.33\%$ (1:12), cross-slope $\leq 2\%$, width $\geq 36"$ (48" PROWAG); sidewalk running slope $\leq 5\%$, clear width $\geq 48"$ PROWAG. Source: U.S. Access Board PROWAG / ADA Chapter 4.

5 · Why it fits this room

City of Boston (the CIO judge)

Boston's CIO runs DoIT on a mission of *government efficiency, resident trust, equity, and human-centered design*, with a planning/housing background and a record of tools for underserved residents. Passable is squarely that: a dignity-first, equity tool that makes a legally-required government function efficient and adoptable tomorrow. He is, literally, the buyer.

Cyvl

Uses the deepest layer of their stack — the new Spatial SDK, calibrated imagery + LiDAR + projection — not just feature records. Demonstrates a vertical their data uniquely enables.

Autodesk

The failing-ramp work-list and 3D segments export to CAD — the existing-conditions handoff a redesign team actually uses.

On NVIDIA's absence

With no model-training judge, Passable's intelligence is *geometry, not a fragile fine-tune* — which de-risks the build. (SAM3 helps locate ramps but isn't load-bearing; failing-ramp locations can be seeded from the existing asset layer.)

6 · The 60-second demo

Hook: "Every city is legally required to know if its ramps are usable. None of them can measure it at scale. We can."

The moment: open a Somerville intersection → type "curb ramp" → the SDK finds it, drops the measured running slope on screen — **9.4%, FAIL** — beside the photo. Pan to the citywide map: green/amber/red ramps, a counter of how many fail, ranked worst-first. Click "export" → the Transition-Plan work-list downloads.

Close: "Somerville's accessibility survey was done by volunteers. This does the whole city from the desk, with a number and a photo behind every verdict — and it re-runs itself every time the street is rescanned."

7 · Saturday scope & honest risks

8-hour build

- **Core (must):** SDK installed → measure slope/width on a handful of known ramps → threshold verdict → map of one neighborhood with pass/fail + photos. This alone is the demo.
- **Then:** citywide ramp run, ranked work-list, export.
- **Stretch:** `locate()` text-prompt detection so it generalizes beyond seeded assets; sidewalk-width pass.

Risks to clear first — verify in the first 30 minutes:

- **The SDK is untested.** Confirm `pip install` + auth works, and that you can pull one real ramp's slope, before committing the day.
- **`locate()` needs a fal.ai API key** (SAM3). If unavailable, seed from the 748 known ramp locations — the measurement still works without it.
- **LiDAR streams over HTTP** — wifi-dependent. Cache the demo neighborhood's tiles locally early so the demo survives a dead network.
- **Measurement accuracy** on a ~5-foot ramp depends on point-cloud density — validate one ramp against a believable hand estimate before trusting the pipeline; show confidence honestly.

What carries over from prior prep: Somerville data knowledge, MCP/API familiarity, the rubric playbook, role split, and the pitch discipline. The data is the same city; only the analysis layer is new.